

BATTERY-POWERED NIGHT LAMP

Using An Old LED Bulb

A. SAMIUDDHIN

This circuit utilises a low-cost 3W LED board to construct a battery-powered night lamp. This board consists of eight SMD2835 LEDs connected in series.

Branded LED bulbs have SMPS drivers and protection circuits for proper illumination. But some cheap LED bulbs have only capacitor power supplies and no protection circuits. In this project, 78S40 IC is used, which acts as a switching regulator system.

SMD2835 LED module. SMD2835

is a novel packaging structure with self-heat-sink design and high-efficiency LED module, widely used in the fields of general, commercial, industrial, city and automotive lighting, among others. Some of its features are given below.

Vertical structure. Chip direct conduction fins improve heat dissipation performance, which can withstand higher current (40-60mA).

Large heat-sink. Heat-sink is 2-3 times larger than SMD3528 with an

excellent heat dissipation design.

Large light-emitting surface. This improves the efficiency of light up to 90 per cent.

Cost. Although SMD2835 is 25 per cent cheaper than SMD3528, the same luminous flux can be achieved using the former.

The circuit shown in Fig. 1 is built around 78S40 DC-DC converter IC (IC1), eight high-bright SMD2835 LEDs (from an old LED bulb) and a few other components. Here, IC1 is configured as a modified boost converter. It acts as both boost converter and constant current driver.

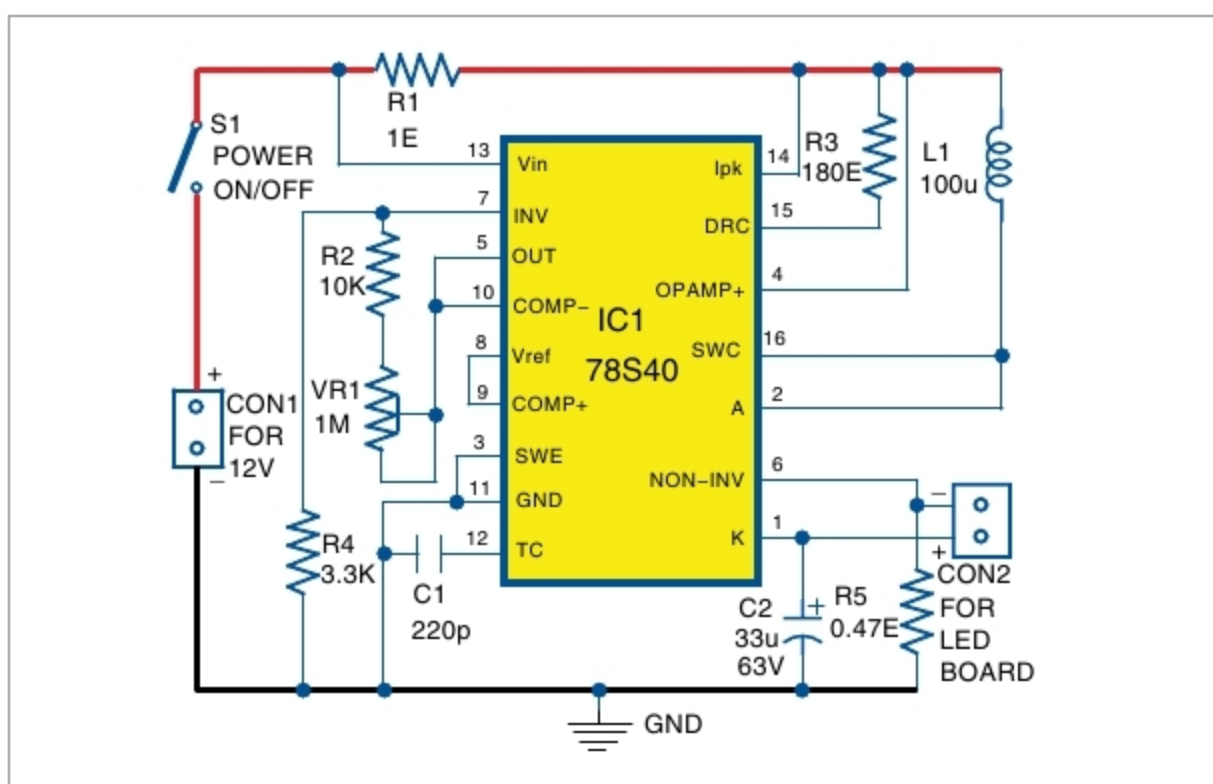


Fig. 1: Circuit diagram of battery-powered night lamp using an old LED bulb

PARTS LIST

Semiconductors:

IC1 - 78S40 DC-to-DC converter

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1	- 1-ohm
R2	- 10-kilo-ohm
R3	- 180-ohm
R4	- 3.3-kilo-ohm
R5	- 0.47-ohm
VR1	- 1-mega-ohm preset

Capacitors:

C1 - 220pF ceramic disk
C2 - 33 μ F, 63V electrolytic

Miscellaneous:

- CON1, CON2 - 2-pin terminal connector
- S1 - On/off switch
- L1 - 100 μ H inductor
- 12V battery
- LED board

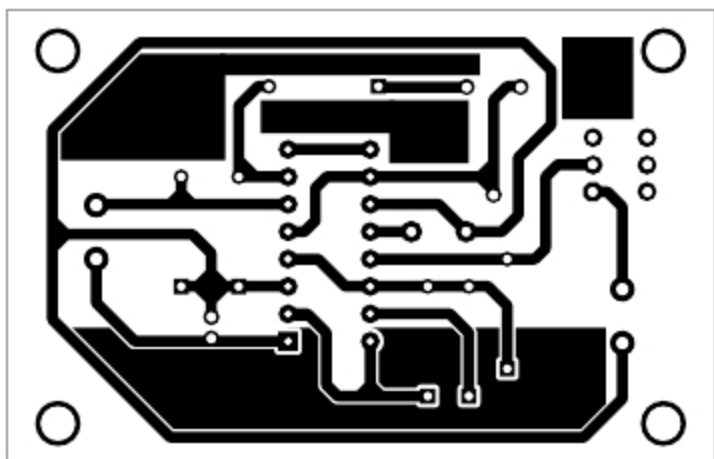


Fig. 2: Actual-size PCB layout of the battery-powered night lamp

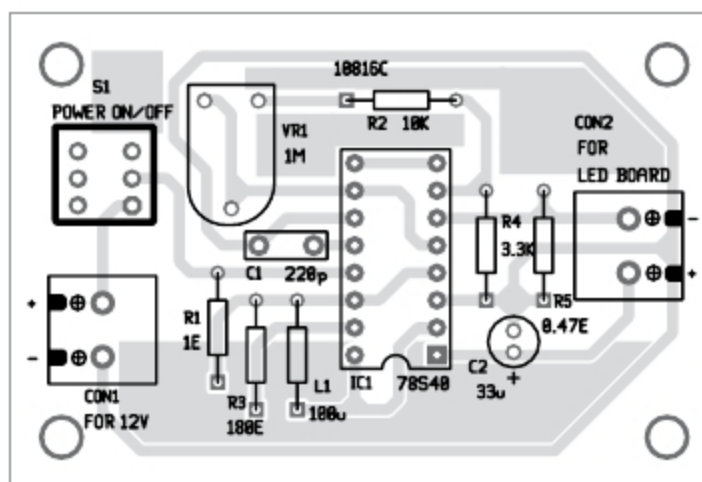


Fig. 3: Components layout for the PCB

IC1 is in a 16-pin dual inline package that allows reference and non-inverting input of the comparator to be pinned out. These additional features greatly enhance the flexibility of this part and allow the implementation of more sophisticated applications.